LAB 01

LLMs, RAG, and Agentic AI

Step by Step Practical Manual

Step 0 Requirements

Students must have access to the following:

- A laptop with Windows operating system
- Internet connection for installation and setup
- Basic familiarity with typing and command entry

Step 1 Install Python

Open a web browser such as Google Chrome or Microsoft Edge.

Navigate to the Python official website at
https://www.python.org/downloads

Click on Download Python.

When the installation window appears, ensure that the option to add Python to PATH is selected. This step is essential for proper execution of Python commands.

Click on Install Now and wait for the installation to complete.

To verify installation, open Command Prompt by pressing the Windows key, typing cmd, and selecting Command Prompt.

Type the following command:

python --version

The system should display the installed Python version.

Step 2 Open PowerShell

Press the Windows key and type PowerShell.
Select Windows PowerShell.

All further commands in this lab will be executed in this terminal.

Step 3 Create Project Folder

In PowerShell, enter the following commands:

```
cd Documents
mkdir ai-interview-assistant
cd ai-interview-assistant
```

Confirm that the current directory path reflects the created folder.

Step 4 Install Streamlit

In PowerShell, type:

```
pip install streamlit
```

To verify installation, type:

```
python -m streamlit --version
```

Step 5 Install Ollama

Open a browser and go to
https://ollama.com/download

Download and install the software for Windows.

After installation, close the current PowerShell window and open a new one.

Verify installation by typing:

```
ollama --version
```

The system should display the installed version.

Step 6 Download the LLM Model

In PowerShell, type:

```
ollama run llama3
```

The model will begin downloading. This may take several minutes depending on internet speed.

Once the process completes and a prompt appears, press Control and C to exit.

Step 7 Create Application File

In PowerShell, type:

notepad app.py

If prompted, confirm creation of the file.

Step 8 Write the Code

Paste the following code into the file:

```
import streamlit as st
import subprocess

st.title("AI Interview Assistant")

role = st.text_input("Enter Job Role", "Data Analyst")
context = st.text_area("Paste Resume or Job Description")

def ask_ollama(prompt):
    result = subprocess.run(
        ["ollama", "run", "llama3"],
        input=prompt,
        text=True,
        capture_output=True
    )
    return result.stdout

if st.button("Generate Question"):
    prompt = f"""
    You are an interviewer.

    Based on this context:
    {context}

    Ask one interview question for {role}.
    """

    question = ask_ollama(prompt)
    st.session_state["question"] = question
```

```
    st.write("Question")
    st.write(question)

if "question" in st.session_state:

    answer = st.text_area("Your Answer")

    if st.button("Evaluate Answer"):

        eval_prompt = f"""
        Question: {st.session_state['question']}
        Answer: {answer}

        Provide feedback and ask a follow up question.
        """

        feedback = ask_ollama(eval_prompt)

        st.write("Feedback")
        st.write(feedback)
```

Save the file and close the editor.

Step 9 Run the Application

In PowerShell, type:

python -m streamlit run app.py

A browser window will open automatically with the application.

Step 10 Test the Application

Enter the job role as Data Analyst.

Provide a sample resume or job description such as experience in Python, SQL, and Machine Learning.

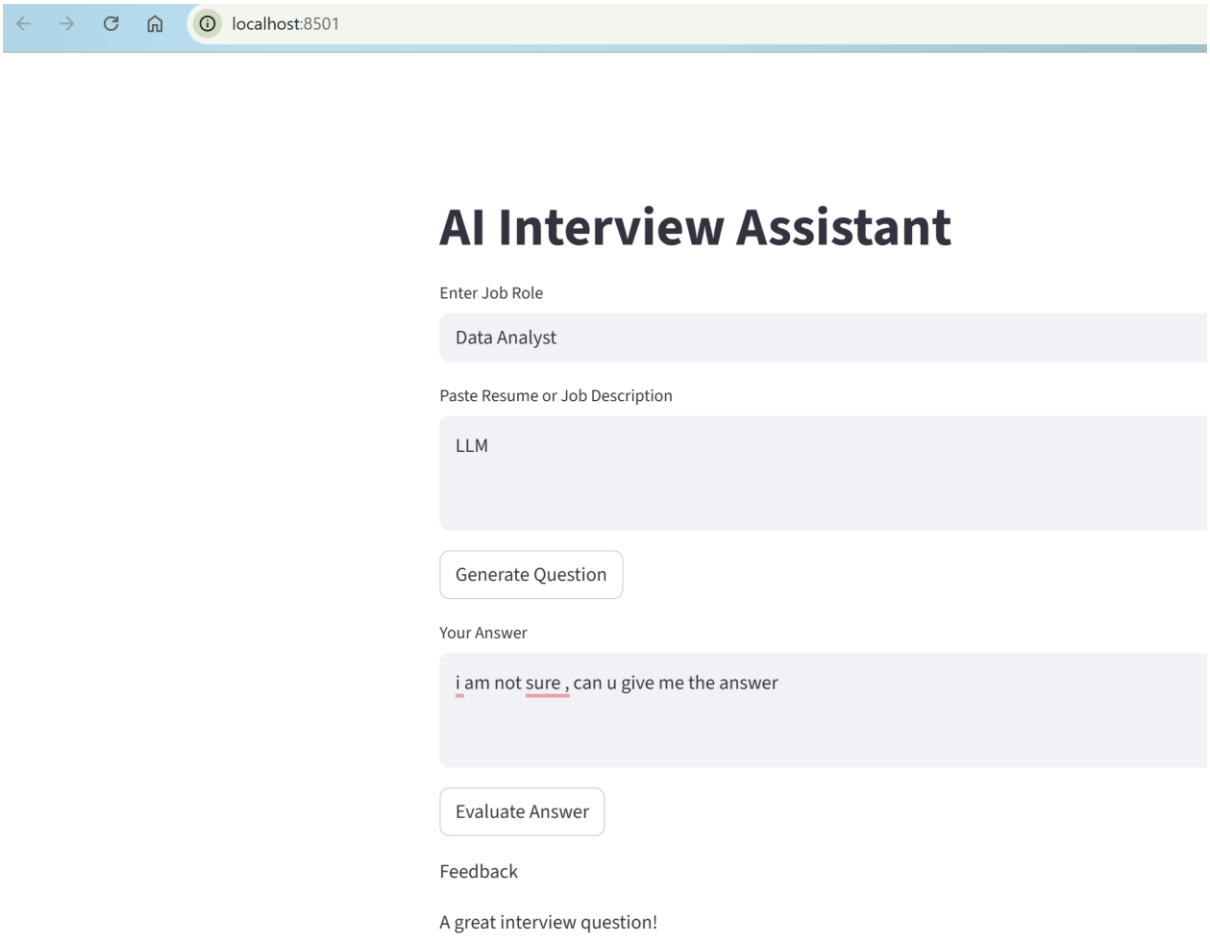
Click Generate Question.

Enter a response and click Evaluate Answer.

Expected Output

The system should display:

- An interview question
- Feedback on the answer
- A follow up question



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Here's an example answer:

"When evaluating data for business decisions, I believe that the most impor[5D[K important aspect is data quality and relevance. This includes factors such [K as data accuracy, completeness, timeliness, and consistency. Without high-q[6D[K high-quality data, any analysis or insights generated from it may be flawed[6D[K flawed, leading to poor decision-making.

To ensure my analysis takes into account potential biases or limitations of[2D[K of the data, I follow a few key steps. Firstly, I thoroughly review the dat[3D[K data's sources, methodology, and assumptions to understand its strengths an[2D[K and weaknesses. This helps me identify potential biases or errors that coul[4D[K could impact the results. Secondly, I perform exploratory data analysis to [K visualize the data and detect any anomalies or irregularities. Finally, I u[1D[K use statistical techniques, such as sensitivity analyses or robustness chec[4D[K checks, to verify my findings and ensure they're not driven by outliers or [K other limitations.

For instance, if I'm analyzing customer purchase behavior, I'd want to cons[4D[K consider factors like seasonality, promotions, or changes in market conditi[7D[K conditions that could influence the results. By being aware of these potent[6D[K potential biases and taking steps to mitigate them, I can increase the conf[4D[K confidence in my findings and make more informed business decisions."

Feedback:

- You showed a good understanding of the importance of data quality and rel[3D[K relevance.
- Your answer was clear and concise, making it easy to follow your thought [K process.
- You provided specific examples and techniques to address potential biases[6D[K biases or limitations, which demonstrates your analytical skills.

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Follow-up question:

"Can you walk me through a time when you had to deal with biased or limited[7D[K limited data? How did you identify the issue, and what steps did you take t[1D[K to correct it?"

This follow-up question will help us dig deeper into your experience and de[2D[K decision-making process in dealing with imperfect data.

Conceptual Understanding

This lab demonstrates three key concepts.

Large Language Models are used through a locally running model to generate responses.

Retrieval Augmented Generation is implemented by providing contextual input such as a resume to guide question generation.

Agentic AI is demonstrated through a multi step workflow that includes question generation, answer evaluation, and follow up interaction.

Troubleshooting

If the file is saved incorrectly with an additional extension, rename it to ensure it ends with .py

If the application is not found, ensure that the correct directory is selected using the change directory command.

If Streamlit does not run, execute the application using the full Python module command.

If Ollama does not respond, restart the terminal and try again.

Final Outcome

By completing this lab, students learn how to build a simple AI application, use a local language model, incorporate contextual reasoning, and design a multi step intelligent workflow.

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